

Sanjoy Kumar Das

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Professional Summary

Software Engineer with strong experience in Python, Node.js, and React, focused on building scalable web applications and AI-powered systems. Experienced in backend API development, cloud platforms, and modern web technologies, with hands-on experience integrating AI models and deploying applications in cloud environments. Passionate about learning new technologies, solving real-world problems, and contributing to fast-growing startup environments.

Technical Skills

Languages: Python, JavaScript, TypeScript, PHP, SQL, Bash

Backend Development: Flask, FastAPI, Node.js, REST APIs, Microservices

Frontend Development: React.js, Next.js, HTML5, CSS3, Vite

Cloud Platforms: AWS, Google Cloud Platform (GCP), Azure

Databases: PostgreSQL, MySQL, MongoDB, MariaDB

DevOps & Tools: Docker, Containerized Applications, GitHub, Jenkins, Postman, Chrome DevTools

AI / Machine Learning: LLM Applications, OpenAI API, LangChain, Hugging Face, Vector Databases, AI Integration

Big Data: Hadoop, Spark, Hive, Elasticsearch

System Design & Architecture

- Distributed systems fundamentals
- Microservices architecture
- REST API design and backend pipelines
- Scalable database design
- Containerized cloud deployments

Professional Experience

Full Stack Developer

PledgX Inc. | 2022 – Present

- Developed and maintained PledgXQualify, a cloud-based SaaS platform that automates contractor qualification workflows for construction companies, lenders, and project owners.
- Designed and implemented scalable REST APIs and backend services using Python and Flask to support contractor data processing, document management, and qualification reporting.
- Built full-stack features using React.js and modern JavaScript frameworks, improving user workflows for contractor onboarding, document submission, and project qualification processes.
- Integrated AI-powered data insights using Google Doc AI, OpenAI APIs and vector databases to assist with contractor analysis and automated recommendations.
- Designed modular backend architecture and microservices, enabling efficient data exchange between multiple stakeholders in the construction ecosystem.
- Improved application reliability and deployment processes through cloud infrastructure management and CI/CD pipelines on AWS and Google Cloud Platform.
- Collaborated with product and engineering teams to deliver new features in fast development cycles, supporting continuous improvement of the platform.

Data Analyst | Database Administrator

Teletalk Bangladesh Ltd. | 2008 – 2020

- Built automated scripts using Python and web scraping tools to collect and process structured datasets from web sources.
- Designed and optimized SQL queries and ORM models for efficient database operations and reporting systems.
- Implemented structured data pipelines for CSV processing and database integration.
- Managed enterprise-level databases, performance tuning, and security.

Projects

AI Car Assistant Platform

Technologies: React, TypeScript, FastAPI, PostgreSQL, Redis, Docker, OpenAI API

- Built an AI-powered assistant that diagnoses car issues and predicts maintenance needs using machine learning APIs.
- Designed backend architecture supporting REST APIs, authentication, and scalable database storage.
- Implemented AI-driven recommendations and automated maintenance workflows similar to modern automotive service platforms.
- Created a responsive UI allowing users to store vehicle data, ask questions, and track maintenance history.
- Integrated Redis caching to significantly reduce API response latency and improve performance of AI responses.
- Added vector search pipelines for automotive knowledge retrieval.

GitHub Link: <https://github.com/sanjoy-kumar/ai-car-assistant>

Production Software

PledgXQualify Platform | <https://qualify.pledgx.com>

- Built and maintained a cloud-based contractor qualification platform used by construction companies, lenders, and project owners to streamline pre-qualification and bonding workflows.
- Enabled contractors to generate qualification statements and submit project documentation digitally, reducing manual paperwork and processing time.
- Contributed to development of an AI-driven contractor performance insight system supporting risk evaluation and project decision making.
- Built with React, Flask, MySQL databases, and cloud infrastructure.

Education

Master of Engineering (MEng) in Artificial Intelligence

Toronto Metropolitan University

Diploma in Full Stack Web Development

Lighthouse Labs, Toronto

Bachelor of Science in Computer Science & Engineering

Dhaka University of Engineering & Technology (DUET)